Scalable Inversion of Contests with Correlated Performances, Including Softmax and Multinomial Probit

Peter Cotton*

August 2026

Abstract

Multinomial probit choice probabilities over n alternatives are Gaussian orthant integrals, computed by simulation for thirty years, one expensive integral per alternative. We observe that the choice event is one-dimensional: whoever wins, wins at some value, so a single shared survival field prices all n alternatives at once. For the covariances applied work actually uses (factor, block, nested, hierarchical: conditional independence given a few shared variables), the field yields all n probabilities in $O(nLQ)$ operations over a lattice of L winning values and Q factor quadrature nodes, and inverts them back to utilities under a fixed, known covariance. For the factor grammar the same pass gives matrix-free Jacobian–vector products and any specified single-removal counterfactual at the same order; the complete removal table has quadratic output size. The Jacobian is a weighted graph Laplacian whose edge weights are photo-finish tie densities: the geometry of substitution between alternatives, from the same pass that prices them. The cleanest evidence is internal: the identical factor conditioning computed one alternative at a time, on the same nodes and lattice, agrees to 3×10^{-15} and costs six hundred times more. Against the field standard, the Geweke–Hajivassiliou–Keane (GHK) simulator at $n = 200$ is two hundred times slower at 10^4 draws, with the lattice's error below the comparison's resolution. Forward pricing is benchmarked to ten million alternatives, inversion to one million. Softmax, Thurstone–Mosteller, factor probit, and team and sector hierarchies are the same two calls, and the same objects serve selection and routing among correlated populations of models or agents.

1 Introduction

A decision maker faces n alternatives and chooses the one with the highest utility. Write $U_i = \mu_i + \varepsilon_i$ with $\varepsilon \sim N(0, \Sigma)$; the probability that alternative i is chosen is

$$p_i \;=\; \Pr\{U_i \geq U_j \text{ for all } j\} \;=\; \Pr\{\tilde{U} \geq 0\}, \qquad \tilde{U}_j = U_i - U_j, \tag{1}$$

an $(n-1)$-dimensional Gaussian orthant integral. This is the multinomial probit model. Its appeal has been understood since Thurstone (1927): unlike the logit family it imposes no independence of irrelevant alternatives, because Σ lets alternatives be close substitutes. Its

*`peter.cotton@microprediction.com`. The reference implementation (Python), its parity-locked ports (R, Rust, JavaScript) and the seeded scripts behind every number in this paper live at `github.com/microprediction/winning`; this revision's tables were produced at repository tag `paper-r3`, package version `1.3.0`.

curse has been understood just as long. The integral (1) has no closed form, and estimation needs it, and its derivatives, at every trial value of the parameters.

The workhorse answer is the GHK simulator of Geweke (1989), Börsch-Supan and Hajivassiliou (1993), Keane (1994) and Hajivassiliou et al. (1996). GHK factors Σ by Cholesky and writes the orthant probability as a nested sequence of one-dimensional truncated normal probabilities, each conditioning on draws for the previous coordinates. Averaging R such importance weights gives an unbiased, smooth estimate of one p_i at cost $O(n^2)$ per draw. The estimator made maximum simulated likelihood practical and it remains the default in textbooks (Train, 2009). Its costs are also standard: error shrinks as $O(R^{-1/2})$; the log of a simulated probability is biased at $O(1/R)$; gradients inherit the simulation noise; and a full probability vector costs n separate runs. A single likelihood contribution needs only the chosen alternative's probability, so the n-fold cost falls not on the individual log-likelihood but on everything else estimation touches: share inversion, aggregate-share and market models, prediction and welfare across the menu, and the derivative of any of these with respect to all n utilities.

This paper takes a different route to the same object, in a min-wins convention fixed here once: performances are times, $X_i = -U_i$, and the lowest time wins. The choice event is one-dimensional. Contestant i wins when its time beats the field, so

$$p_i = \int f_i(x) \prod_{j \neq i} S_j(x)\, dx, \tag{2}$$

where f_i is the density of X_i and S_j the survival of X_j: integrate over the winning time x, requiring every rival to come in slower. For independent performances Cotton (2021) computed (2) for all n on a lattice by building the field $\sum_j \log S_j$ once and dividing each alternative out, and inverted the map from probabilities back to μ. The present paper extends the field construction to correlated performances by conditioning: whenever Σ makes performances independent given a few shared variables, the field factorizes and all n probabilities cost $O(nLQ)$ for a lattice of L points and Q quadrature nodes. For the factor grammar the derivatives ride the same pass at the same cost: Jacobian–vector products and each inversion iteration are $O(nLQ)$, with the Jacobian a weighted graph Laplacian applied as an operator and never materialized. The block and nested kernels materialize their Jacobians, and the tree's cross-cluster block is approximate, so the matrix-free claim is the factor grammar's, not every grammar's. No per-alternative repetition; error controlled by quadrature, deterministic for the Gauss–Hermite rules and reproducibly randomized for scrambled-Sobol nodes.

Only utility contrasts and a normalized scale are identified in multinomial probit, so the raw covariance is overparameterized, and the contrast covariance may in principle carry unrestricted parameters. Applied work nevertheless imposes structured covariance as a matter of course, for parsimony and for computation (Train, 2009); the grammars here cover the structured families in common use. The table lists them: the first six rows are nested exactly and each is priced and inverted by the same two calls; the last row is a fitted approximation developed in §6.

model	covariance grammar	source
Luce / softmax	independent, minimum-Gumbel base, $D = \pi^2/6$	exact identity
Thurstone–Mosteller	independent, normal base	Cotton (2021)
factor multinomial probit	$VV' + \mathrm{diag}(D)$	§4.1
teams, conferences	blocks: rank-r effect per cluster	§4.2
market plus sectors	nested: blocks $\times$ global factor	§4.2
hierarchies	tree: uniform shared effects	§4.3
dense Σ	fitted grammar + residual factors (approx.)	§6

The second motivation is scale. Measured, on one laptop: all-n forward pricing at $n = 10^4, 10^5, 10^6$ in 0.18, 2.7 and 29 seconds (rank-one factor, Rust kernels), a ten-million-contestant block field in 245 seconds, and inversion at $n = 10^6$ in 80 seconds (independent) and 22 minutes (rank-one factor). GHK's measured complete-vector cost is superquadratic and steepening (log-log slope 2.15 from $n = 10$ to 200, 2.8 from 200 to 1000; timings at $R = 10^3$ of 0.57, 5.66 and 51.3 seconds at $n = 200, 500, 1000$, scaling by ten at $R = 10^4$), so the gap widens without bound; what the cost law implies at 10^4–10^6 alternatives is an extrapolation and is deferred to Appendix A. Every timing is regenerated by the committed `bench.py`.

Section 2 places the method among the alternatives. Section 3 gives the independent lattice race; Section 4 extends it across the covariance grammar, factor to blocks to trees, each with its recursion written out. Section 5 gives the Jacobian operator and the inversion. Section 6 reduces dense covariance to the grammar family, approximately, and reports accuracy per named random-correlation ensemble. Section 7 contains the benchmarks.

2 The alternatives

Deterministic quadrature. Genz (1992) transformed the orthant integral to the unit cube and integrated by adaptive and quasi-Monte Carlo rules; the `mvtnorm` implementation is the standard in R. Accuracy is not the issue: in the benchmark below it agrees with the lattice to the reference's own noise. The shape is the issue. It treats one probability at a time, so the all-n vector at $n = 30$ costs three hundred times the lattice, and the gap widens with n.

Frequency simulation. The crude frequency estimator of Lerman and Manski (1981) counts argmax draws. It is unbiased for the whole vector at once but not smooth in parameters, and its per-entry relative error explodes for small p_i. Measured below: at wall clock matched to the lattice it carries twelve to twenty times the total-variation error, records zero counts on tail alternatives at $n = 30$, and at ten times the budget remains a factor of two to four behind.

GHK. Defined above. Its dominance is documentable rather than folklore. Stata's `cmmprobit` manual states that the likelihood evaluator "implements the Geweke–Hajivassiliou–Keane (GHK) algorithm to approximate the multivariate distribution function" (StataCorp, 2023; Gates, 2006), and the command's entire `intmethod` menu (Hammersley, Halton, or pseudo-random point sets) selects only the sequence feeding the same simulator: the market-leading implementation offers no non-GHK method at all. R's `mlogit` and `mvProbit` default to GHK likewise (Croissant, 2020), and the GHK-based `mvprobit` of Cappellari and Jenkins (2003) carried the method to applied Stata work at large. Two properties matter for comparison: it is smooth in parameters, which frequency counting is not, and it prices one alternative per run. Stern (1992) gave a related decomposition with an analytic leading term.

Bayesian data augmentation. Albert and Chib (1993) and McCulloch and Rossi (1994) sidestep the integral entirely by Gibbs sampling the latent utilities; Imai and van Dyk (2005) implement the marginal data-augmentation version in the `MNP` package, and Loaiza-Maya and Nibbering (2022) carry the Bayesian route to large choice sets by variational methods. For Bayesian inference this is a complete and often excellent answer. It is simply answering a different question than the one benchmarked here: the cost moves into the Markov chain, the likelihood surface is never formed, and choice probabilities appear only as posterior functionals, so there is no pricing call to time or to check.

Analytic approximations. Clark (1961) matched moments of pairwise maxima; Mendell and Elston (1974) and Solow (1990) approximate by sequential conditioning on first and second moments. These are fast, sometimes startlingly accurate, and carry no error control, and the benchmark below measures all three properties at once: Mendell–Elston prices the $n = 10$ vector in two milliseconds within 2×10^{-3} of the reference, then drifts to 3×10^{-2} at $n = 30$ with tail probabilities off by a factor of nearly four, and nothing in the output announces the change. The Clark-type recursion behaves alike. The family is current practice, not history: Bhat (2011) builds the MACML estimator on precisely these approximations, and the `Rprobit` package implements it (Bauer, 2024), with its asymptotics examined by Batram and Bauer (2019). Expectation propagation is the modern member of the same family. We benchmark representative sequential moment approximations through the classical members; MACML's composite-likelihood construction around its analytic ingredient is its own estimator and is not itself benchmarked here.

Minimax tilting. Botev (2017) computes single Gaussian rectangle probabilities with bounded relative error deep into the tails, in dimensions up to roughly a thousand. It is the accuracy reference, and we use it as the referee for our own tail claims; it prices one probability per run.

Every smooth, likelihood-grade method above prices one probability at a time; frequency counting obtains the vector but not derivatives. A single likelihood contribution is content with one probability; the tasks that surround it are not. Share inversion, prediction across the menu, welfare, and derivatives with respect to all n utilities each want the whole vector at every parameter value. Sharing the field across alternatives is the algorithmic point of this paper, and it is what makes the n-fold and the derivative costs disappear for those tasks.

3 The lattice race

In the min-wins convention of the introduction, let $X_i = \mu_i + \sigma_i \epsilon_i$ with ϵ_i i.i.d. from a standardized base density (normal, Gumbel, or any density supplied as code), and write f_i, S_i for the density and survival of X_i. From here on μ_i is a *time* location, the negative of the introduction's utility location: raising it makes contestant i worse and lowers p_i. Inversion returns time locations, and utility coefficients are their negation.

The algorithm evaluates (2) on a lattice $x_1 < \cdots < x_L$:

1. compute $\log S_j(x_\ell)$ and $\log f_j(x_\ell)$ for all j, ℓ;

2. accumulate the field $L(x_\ell) = \sum_j \log S_j(x_\ell)$;
3. for each i, form the integrand $f_i(x_\ell)\, e^{L(x_\ell) - \log S_i(x_\ell)}$, which is contestant i's density times everyone else's survival, with i divided out of the shared field in log space;
4. sum times the lattice spacing, and normalize the vector.

The cost is $O(nL)$: the field is built once, not once per contestant. Log-space arithmetic keeps hopeless contestants finite (their $e^{L-\log S_i}$ underflows gracefully rather than dividing zero by zero), and the exponent is clipped at -745.

The lattice itself is placed on the winner's bulk, not the ability span. A hopeless contestant wins only by running a winner-class time, so only the region where the winning time lives matters. Bisect the conservative winner c.d.f. envelope $G(x) = 1 - \prod_j S_j(x)$, built from the caller's own base survival, to $[G^{-1}(\delta), G^{-1}(1-\delta)]$ and pad two standard deviations, which makes the truncated tail mass negligible against the accuracy floors reported below. Two details make that description literally true rather than nearly true. The interval is bracketed by geometric expansion before it is bisected: nine standard deviations hold the δ-quantiles of a Gaussian race but not of an arbitrary polynomial tail, and bisecting an interval that does not bracket the root converges to its endpoint while appearing to succeed. And δ is a request rather than a guarantee, because the two error sources trade against each other: a t_4 race puts its 10^{-12} winner quantile 1456 standard units out, and honoring that at 257 points would spread the lattice to a spacing of 5.7 and lose far more to discretization than truncation ever cost. When the requested window will not fit the point budget at a spacing of half the tightest performance standard deviation, δ is relaxed by factors of one hundred until it does, and the achieved value is reported. The window is then exactly the stated construction at the δ it names. The envelope must use the supplied base, not a normal surrogate: a Student-t base at $\nu = 2.5$ has polynomial tails that a normal envelope clips, costing 5×10^{-3} of total variation at $n = 40$ against 5×10^{-5} once the true survival sets the window. Measured: 33 points on this window reach accuracy that a 500-point ability-span lattice cannot, because the span window's own truncation floors near 6×10^{-11}; on fields with many hopeless contestants the bulk window is one hundred times more accurate at equal budget.

The same lattice carries the derivatives. A Jacobian evaluated on a different grid, or one that treats the lattice total as stationary when it is not, is the derivative of a neighbouring map rather than of the one that was evaluated; both matter at the accuracy this paper claims. The shipped full and row Jacobians therefore build the window through the same routine as the forward pass, integrate the own-coordinate derivative $\partial f_i/\partial \mu_i$ instead of imposing it from the continuum's zero-row-sum identity, and apply the quotient rule to the lattice normalization: with unnormalized masses a, total $T = \mathbf{1}^\top a$ and $p = a/T$, the returned matrix is $(A - p\,\mathbf{1}^\top A)/T$ with $A = \partial a/\partial \mu$. Against central differences of the forward map this is exact to 2×10^{-11} relative on a coarse 65-point lattice and 1×10^{-6} under a t_4 base, where the previous continuum surrogate stood at 1×10^{-8} and 1×10^{-5}; on a fine Gaussian lattice the two agree, which is why the surrogate survived earlier scrutiny. Row sums now vanish as a property of the quadrature rather than by construction.

Two byproducts are cheap once the field is built. Any specified photo-finish tie density is one more lattice sum; §5 identifies it with the off-diagonal Jacobian. A removal counterfactual, the probability j wins after deleting i, is one more division in the continuum, but the *numerical* field must be rebuilt on a window that covers the post-removal winner: delete a dominant favorite and the surviving race lives in the original race's second-order-statistic region, where the winner-bulk lattice carries no mass at all. With $\mu = (-20, 0, 1)$ the bulk window is about $[-29, -11]$, the post-removal truth is $\Phi(1/\sqrt{2}) = 0.760$, and dividing the favorite out on the original grid leaves 3×10^{-28} of unnormalized mass, whose renormalization is numerical debris. The implementation therefore integrates removals on their own span window and checks the unnormalized mass, raising on a defect rather than renormalizing it away. The full $n \times n$ matrices of such queries are $\Omega(n^2)$ outputs and are priced per query, not free.

4 The covariance grammar

Three structures make contestants conditionally independent given a few shared variables, in increasing order of hierarchy: a global factor, factors per cluster, and a tree of uniform shared effects. Each keeps the shared field and the $O(nLQ)$ bill; each is exact, not fitted. The approximate reduction of a dense covariance to this family is a separate matter and has its own section (§6).

4.1 Factor

Let $X = \mu + Vf + \mathrm{diag}(\sqrt{D})\,\epsilon$ with $f \sim N(0, I_k)$ independent of ϵ, so $\Sigma = VV' + \mathrm{diag}(D)$. Conditional on f the performances are independent with means $\mu + Vf$, and

$$p \;=\; \sum_q w_q\, p^{\mathrm{ind}}(\mu + Vf_q), \tag{3}$$

a Q-node quadrature over runs of the independent algorithm, cost $O(nLQ)$. In English: average the independent race over the shared luck. The node rule is adaptive rather than fixed by rank: a tensor Gauss–Hermite grid while it stays under 10^5 nodes, so ranks three and four still use it; scrambled Sobol once the integrand is sharp or the tensor would exceed that budget; and an equal-weight midpoint-quantile grid at extremely sharp rank one, where Gauss–Hermite's clustered abscissae are the wrong family for a near-step integrand. Conditioning on the factor is not new: Butler and Moffitt (1982) built exactly this quadrature for the one-factor random-effects probit, and the `lpRR`/`slpRR` interfaces of `mvtnorm` integrate over the factor dimensions of $BB' + \mathrm{diag}(D)$ by Monte Carlo. What those constructions price is one alternative per integral. The field is the new part: one pass shares the survival product across all n alternatives, and §7 measures the difference against the per-alternative version of this same quadrature.

One default matters. When $D_i \ll \|V_i\|^2$ the conditional race is nearly deterministic and the integrand over f is nearly a step function; a fixed 15-node rule then silently loses up to five percent of total variation, and the loss is invisible to lattice refinement because it is points-invariant. The default order now scales with the sharpness $\max_i \|V_i\|/\sqrt{D_i}$, capped by rank.

The same pass returns the own-parameter slopes $\partial p_i/\partial \mu_i$ at no extra cost (negative in this min-wins convention: raising a time mean lowers the win probability), which §5 uses as the Newton preconditioner.

4.2 Blocks and nested

Assign contestant i to cluster $c(i)$ with loading $v_i \in \mathbb{R}^r$ and let

$$X_i = \mu_i + v_i' a_{c(i)} + \sqrt{D_i}\,\epsilon_i, \qquad a_c \text{ i.i.d. } N(0, I_r), \tag{4}$$

a block-diagonal rank-r-plus-diagonal covariance: teams, conferences, sibling products. Two independences do the work. Conditional on its own cluster's effect a contestant is independent of its teammates; and distinct clusters are independent outright. So the joint survival factorizes across clusters,

$$\Pr\{X_j > x \text{ for every } j\} \;=\; \prod_c G_c(x), \qquad G_c(x) \;=\; \mathbb{E}_a \prod_{j \in c} S_j(x \mid a) \;=\; \sum_q w_q\, e^{S_c(x, a_q)}, \tag{5}$$

with $S_c(x, a) = \sum_{j \in c} \log S_j(x \mid a)$ the cluster's conditional log-survival. In English: $G_c(x)$ is the probability that every member of cluster c survives past x, with the cluster's shared luck integrated out.

Contestant i then wins against a cavity field:

$$p_i \;=\; \int h_i(x) \prod_{c \neq c(i)} G_c(x)\, dx, \qquad h_i(x) \;=\; \sum_q w_q\, f_i(x \mid a_q)\, e^{S_{c(i)}(x, a_q) - \log S_i(x \mid a_q)}. \tag{6}$$

The term h_i couples i's win density to its own teammates within the shared draw of $a_{c(i)}$; the cavity product supplies the rest of the field. The cost is $O(nLQ_a)$ whatever the number of clusters, because the per-cluster factors are accumulated once. Rank two is special: the conditional integrand is smooth and a tensor Gauss–Hermite rule beats scrambled Sobol at equal nodes by a wide measured margin.

Nested. Add one global factor with couplings g_i and a dial $\gamma \in [0, 1]$: conditional on the global draw f_q the model is a block race at shifted abilities, so $p = \sum_q w_q\, p_{\text{block}}(\mu + \gamma g f_q)$, a finite mixture of block races. At $\gamma = 0$ the clusters are isolated, at $\gamma = 1$ fully coupled, and the covariance contribution scales as $\gamma^2 g g'$.

4.3 Trees

Let a rooted tree have the leaf clusters at its bottom and internal nodes t above, and give node t a scalar effect $b_t \sim N(0, 1)$ applied with uniform strength λ_t to every leaf beneath it:

$$X_i = \mu_i + v_i\, a_{c(i)} + \sum_{t \,\in\, \text{anc}(c(i))} \lambda_t b_t + \sqrt{D_i}\,\epsilon_i. \tag{7}$$

The implied covariance is hierarchical: two contestants share the variance of exactly their common ancestors. The root's effect is common to every contestant, hence choice-irrelevant, and is fixed to zero. The uniform strength is the crucial restriction, because a shared effect that hits every subtree member equally moves the whole subtree's field by a translation of its argument. Messages therefore remain functions of one variable. Per-leaf loadings on internal effects would force two-dimensional message tables.

The algorithm is two passes over the same lattice.

Upward. Each leaf cluster carries its block factor $G_c(x) = \sum_q w_q e^{S_c(x,a_q)}$ from (5). Each internal node, visited deepest first, combines its children under its own effect:

$$G_t(x) \;=\; \sum_q w_q \prod_{c \in \mathrm{children}(t)} G_c(x + \lambda_t a_q)\,. \tag{8}$$

In English: $G_t(x)$ is the probability that every leaf below t survives past x, with t's effect integrated by quadrature and each child's message shifted because the effect moves that child's whole subtree at once. Shifted evaluations are linear interpolations on the lattice.

Downward. The root's cavity is one. Each node, visited shallowest first, receives the field of everything outside its subtree:

$$R_t(x) \;=\; \Big[\sum_q w_q\, R_{\mathrm{parent}(t)}\big(x - \lambda_{\mathrm{parent}(t)} a_q\big)\Big] \prod_{s \in \mathrm{siblings}(t)} G_s(x). \tag{9}$$

The sibling messages sit outside the parent-factor quadrature. This placement is correct but not obvious: the parent effect shifts the candidate and every sibling subtree by the same amount, so it cancels from every candidate-versus-sibling comparison; only the cavity from outside the parent must be shifted and averaged. Contestant i then wins against its own leaf cavity, $p_i = \int h_i(x)\, R_{c(i)}(x)\, dx$ with h_i as in (6). The cost is $O(nLQ)$ plus $O(LQ)$ per tree node.

A depth-one tree with zero strengths reproduces the block race to machine precision, which is tested. Against a 2^{22}-path common-random-numbers referee the recursion sits at the simulation noise floor on every resolvable entry at depths one through four (root-mean-square z between 0.75 and 1.1). Entries below the referee's resolution agree with minimax tilting (Botev, 2017) to 0.3 percent relative at probabilities down to 7×10^{-8}, which is within the tilting estimate's own error. Given adequate factor quadrature, tail accuracy is carried by the lattice, and the grammar kernels price tails that simulation does not resolve.

Remark 1 (A connection to financial practice)**.** *Long-only portfolio weights could be considered as race probabilities, and there is a tree-race covariance under which hierarchical risk parity's bisection geometry (López de Prado, 2016) is exactly a race: give internal node t strength $\lambda_t^2 = \rho_t - \rho_{\mathrm{parent}(t)}$ with $\rho_t = \max(1 - 2h_t^2, 0)$ for merge height h_t, and leaf i idiosyncratic variance $1 - \rho_{\mathrm{first}(i)}$. The shipped* `from_linkage` *retains the root at its cophenetic value, so its implied correlation is the floored cophenetic matrix itself. Fixing the root effect to zero instead gives a contrast-equivalent race, the root being choice-irrelevant: the two agree on every pairwise difference variance, but the zero-root model's raw correlations are $(\rho_{ij} - \rho_0)/(1 - \rho_0)$ rather than ρ_{ij}, which matters only to a reader reading correlations off the model rather than choices out of it. The floor is forced, because uniform shared effects cannot represent negative dependence, and omitting it silently inflates other implied correlations above one. The identity turns dendrogram-consistent concentration limits into contest inversions, but portfolio applications are not this paper's subject.*

5 Jacobians, tie densities, inversion

The Jacobian is a boundary flux, and saying so once derives its formula, its symmetry, its conservation law and its graph structure together. Contestant i wins on the polyhedral cone

$R_i = \{x : x_i \le x_k\ \forall k\}$, so $p_i(\mu) = \int_{R_i} q_\mu$ with $q_\mu(x) = q_0(x - \mu)$; the cones meet along the photo-finish faces $F_{ij} = \{x_i = x_j \le x_k,\ k \ne i, j\}$. Because μ enters only as a translation, $\partial_{\mu_j} q_\mu = -\partial_{x_j} q_\mu$, so for $j \ne i$

$$\frac{\partial p_i}{\partial \mu_j} = -\int_{R_i} \frac{\partial q_\mu}{\partial x_j}\, dx = -\int_{\partial R_i} q_\mu\, n_j\, dS,$$

the second equality being the divergence theorem, the Euclidean special case of the generalized Stokes theorem, applied to the vector field $q_\mu e_j$. Take $q_0 \in C^1$ with q_0 and ∇q_0 integrable and $|x|^{n-1} q_0(x) \to 0$, which the Gaussian law of the theorem satisfies outright; integrability alone would not license either the differentiation under the integral or the flux at infinity. The contribution over the sphere at infinity then vanishes, and the only face of ∂R_i whose outward normal has a nonzero j-component is F_{ij}, where that normal is $(e_i - e_j)/\sqrt{2}$.

Proposition 1 (Photo-finish derivatives are a boundary flux)**.** *For $j \ne i$,*

$$\frac{\partial p_i}{\partial \mu_j} \;=\; \frac{1}{\sqrt{2}} \int_{F_{ij}} q_\mu\, dS \;\ge\; 0, \tag{10}$$

the probability flux across the surface on which i and j dead-heat ahead of the field. Parametrizing that face by the common value $x_i = x_j = t$ contributes a surface factor $\sqrt{2}$ which cancels, giving

$$\frac{\partial p_i}{\partial \mu_j} = \int \varphi_{ij}(t,t)\ \Pr\{X_k > t\ \forall k \ne i, j \mid X_i = X_j = t\}\, dt, \tag{11}$$

and under independence

$$\frac{\partial p_i}{\partial \mu_j} \;=\; \int f_i(x)\, f_j(x) \prod_{k \ne i,j} S_k(x)\, dx. \tag{12}$$

Under conditional independence the same factorization holds node by node and is averaged over the shared variables. Each row of the Jacobian sums to zero.

Everything else follows from the geometry. Symmetry is $F_{ij} = F_{ji}$: one surface, one number, so $w_{ij} = \partial p_i / \partial \mu_j$ is symmetric and nonnegative. The zero row sum is translation invariance, a conservation law: what leaves one cone enters its neighbours. Writing B for the oriented incidence matrix of the photo-finish graph and W_e for the diagonal of edge conductances w_{ij},

$$J = -B^\top W_e B, \qquad (Jh)_i = \sum_{j \ne i} w_{ij}\,(h_j - h_i),$$

the discrete analogue of $-\operatorname{div}(a\nabla h)$: Bh takes differences across photo-finish boundaries, $W_e Bh$ is the flux across them, $B^\top W_e Bh$ the net flow into each winning region. The mass-one identity is the same theorem in dimension one, since $\sum_i f_i \prod_{j \ne i} S_j = \frac{d}{dt}(1 - \prod_j S_j)$ integrates to one; this is why the implementation's unnormalized mass checks are diagnostics rather than cosmetics, a material defect meaning the lattice missed the region, not that normalization was needed.

The Jacobian is applied as an operator: individual entries and Jacobian–vector products are byproducts of the same field pass (per quadrature node a Gram structure over the lattice), while materializing all n^2 entries is a choice with $\Omega(n^2)$ cost that nothing downstream requires. Three derivative objects should be distinguished: the continuum identity (12); its quadrature evaluation; and the exact derivative of the normalized, adaptive-lattice numerical map, which differs from the first two through the normalization quotient and grid motion. The implementation's grid form differentiates the rectangle sum itself and applies the normalization quotient $(v - p\,\mathbf{1}^\top v)/T$, with T the unnormalized total. That quotient is not optional: without it the returned directional derivative differs from central finite differences of the returned map by 2×10^{-3} at $L = 25$. With it, agreement is 3×10^{-11} from $L = 101$ upward, the production range. A residual 6×10^{-4} survives at $L = 25$ because the window itself moves with μ; that is a grid-motion term, which no fixed-grid derivative contains and which a finite difference recomputing its own window will always see. The continuum form, whose symmetry serves preconditioning, is exposed alongside. The stable pair factorization is $[f_i e^{L - \log S_i}] \times [f_j/S_j]$: the first factor is bounded, and the hazard f_j/S_j grows linearly in the normal base and exponentially in the Gumbel base, either of which the exponent clip absorbs. The symmetric square-root split overflows where survivals vanish.

The block Jacobian is exact, with the within-cluster term computed under the cavity; the nested Jacobian is the mixture of block Jacobians; the tree Jacobian is exact within a cluster and Gram-approximate across clusters, which suffices for Newton because residuals always use the exact forward map. Feasibility-critical callers fall back to finite differences.

The inversion rests on a convex program.

Theorem 1 (The inverse exists and is unique on contrasts)**.** *Fix a covariance of full support (any member of the grammar with every $D_i > 0$) and let $W(\mu) = \mathbb{E}\min_i(\mu_i + \epsilon_i)$, $\epsilon \sim N(0, \Sigma)$. Then W is concave and differentiable with $\nabla W(\mu) = p(\mu)$, and its Hessian is $-L(\mu)$, the negative of the graph Laplacian whose edge weights are the tie densities* (12)*. Every tie density is strictly positive, so W is strictly concave on the contrast space $\mathbf{1}^\perp$. For any target with $p_i^\star > 0$ and $\sum_i p_i^\star = 1$, the program*

$$\max_{\mu}\; W(\mu) - \langle p^\star, \mu\rangle$$

is constant along $\mathbf{1}$, strictly concave and coercive on $\mathbf{1}^\perp$, and its unique mean-zero maximiser is the inverse: $p(\mu^\star) = p^\star$. If some $p_i^\star = 0$ the supremum is approached only as the contrast $\mu_i - \min_{j:\, p_j^\star > 0} \mu_j \to +\infty$ (the raw coordinate statement depends on the gauge); no finite inverse exists, and a floored target recovers a lower bound on that contrast.

Proof. A minimum of affine functions of μ is concave and expectation preserves this. Ties have measure zero under full support, so the minimiser is almost surely unique and Danskin's theorem gives $\partial W/\partial \mu_i = \Pr\{i \text{ wins}\}$.

The Hessian is Proposition 1, which assumed no independence: its off-diagonals are the boundary fluxes (11) under the full correlated law, and its rows sum to zero by the same translation invariance $W(\mu + c\mathbf{1}) = W(\mu) + c$ that fixes the diagonal without further computation. The Hessian is therefore $-L(\mu)$ with L the weighted graph Laplacian of the photo-finish graph; under full support ($D_i > 0$, so every pairwise difference has positive variance) each edge weight is strictly positive, the tie graph is complete, and $L \succ 0$ on $\mathbf{1}^\perp$: W is strictly concave there.

Coercivity follows from the elementary bound $W(\mu) \le \min_i \mu_i$: for each i, $\min_k(\mu_k + \epsilon_k) \le \mu_i + \epsilon_i$, and $\mathbb{E}\epsilon_i = 0$. For a nonconstant unit contrast d and interior target, $\langle p^\star, d\rangle > \min_i d_i$, because $p^\star$ puts positive mass on a coordinate where d exceeds its minimum; hence

$$W(td) - t\langle p^\star, d\rangle \;\le\; t\big(\min_i d_i - \langle p^\star, d\rangle\big) \;\longrightarrow\; -\infty$$

linearly in t. The decay is uniform, not merely raywise: $d \mapsto \langle p^\star, d\rangle - \min_i d_i$ is continuous and strictly positive on the compact unit sphere of $\mathbf{1}^\perp$, so it attains a positive minimum κ, and the displayed inequality at $t = \|\mu\|$ gives $F(\mu) \le -\kappa\|\mu\|$ for every $\mu \in \mathbf{1}^\perp$, writing F for the objective — which is coercivity for arbitrary diverging sequences. (The bound is on F itself, not on F relative to $F(0)$: the objective may climb toward its maximizer before the linear decay takes over.) Constancy along $\mathbf{1}$ is translation invariance again, exactly when the target sums to one. A strictly concave coercive function on $\mathbf{1}^\perp$ attains a unique maximum, where the gradient vanishes: $p(\mu^\star) = p^\star$. □

The program is the Legendre dual of the social surplus function of McFadden (1981), and share inversion as a convex problem is established territory: Berry (1994) inverts market shares in logit-family demand, Galichon and Salanié (2022) identify the general random-utility inversion with optimal transport, and Chiong et al. (2016) compute it by linear programming in the discrete case. Closest to an algorithm is Li (2018), who for arbitrary non-atomic bases writes the inversion as an unconstrained convex minimization whose objective gradient is the share map, with globally convergent superlinear methods — the formulation and the optimizer, without a fast evaluator of the shares and their Jacobian at large n. The shared field is that evaluator, so the two are complements. What the field representation adds is not the convex program but its derivatives: the Hessian is the tie-density Laplacian of the same field pass, so gauge-fixed Newton runs matrix-free at $O(nLQ)$ per iteration.

Identification lives on the same contrast space. Choice probabilities are invariant to a common utility shock, so the model depends on Σ only through $P\Sigma P$ with $P = I - \mathbf{1}\mathbf{1}^\top/n$: a factor proportional to $\mathbf{1}$, or a tree-root effect applied to every leaf, is choice-irrelevant. The right fitting objective for a dense covariance is therefore the projected residual,

$$\min_{V,\, D \ge 0} \left\| P\,(\Sigma - VV' - \operatorname{diag} D)\,P \right\|_F^2,$$

not a fit of $VV' + \operatorname{diag}(D)$ *to* $P\Sigma P$ (the projection of a rank-plus-diagonal matrix is no longer rank-plus-diagonal, so the second problem manufactures error the first does not have). The reference implementation minimizes the stated objective (`factor_model_projected`: alternate a top-k PSD step for V in the contrast basis with the exact nonnegative diagonal solve, whose normal equations are $(P \circ P)\, d = \operatorname{diag}(P(\Sigma - VV')P)$), and §6's pipeline consumes only projected residuals downstream, so an input already of the form $VV' + \operatorname{diag}(D)$ is returned undistorted. The introduction's identified-class sentence is to be read in this sense: structured covariance is the estimable and computable restriction on $\mathbf{1}^\perp$, not a claim that structure exhausts identification.

The shipped iteration is stated so it can be checked against the code (`abilities_from_race`).

Inversion (all grammars; min-wins). Given a target $p^\star$ with positive entries summing to one, set $t = \log p^\star$ and initialize $\mu^0 = -(t - \bar{t})/2$. Repeat until $\max_i |r_i| < 10^{-8}$:

1. one field pass: $\hat{p} = p(\mu)$, and own-slopes $s_i = \partial p_i / \partial \mu_i$ (factor grammar: the same pass; block/nested/tree: the slopes of the independent race at matched total variances, one extra $O(nL)$ pass);
2. log residual $r_i = \log \hat{p}_i - t_i$; log-domain slope $d_i = \min(s_i/\hat{p}_i, -10^{-6})$;
3. damped diagonal Newton step $\mu_i \leftarrow \mu_i - \mathrm{clip}(\alpha\, r_i/d_i, \pm 2)$, then recenter $\mu \leftarrow \mu - \bar{\mu}$.

Damping: $\alpha = 1$ (0.7 at $n = 2$, where the undamped update on the K_2 photo-finish graph is a two-cycle); for the structure grammars $\alpha = 0.7$, halved whenever the residual fails to shrink.

This is a diagonally preconditioned Newton iteration on log probabilities; Theorem 1 makes the target a well-posed problem, and the clip and recentering confine the iterate to the contrast space. The theorem establishes well-posedness, not convergence of this clipped iteration; the iteration is an empirically safeguarded solver, and a convex or Newton–Krylov fallback exists in principle should a case demand it. Round trips through the shipped front door at $n = 400$ (the committed `bench.py invert`; twenty clusters, tree of depth three): independent 0.04 s, tree 0.37 s, blocks 0.67 s, factor rank-2 1.7 s, nested 3.7 s, each to a maximum log-probability residual below 10^{-8}. A zero or negative target raises rather than being silently repaired: by Theorem 1 it has no finite inverse. Flooring is opt-in, and the result is then a one-sided bound on the floored contrasts in the direction the theorem states; the returned diagnostics name which entries were floored, whether the iteration converged and the residual reached, and non-convergence warns rather than returning a best effort silently. The contract is the same for every grammar: the target is validated once, before the call splits by structure, and all five paths return through one exit, so a block or tree inversion reports floored entries, iteration count and convergence exactly as the factor one does. Ignoring known structure at inversion time is not a small error: inverting block-generated probabilities under an assumed independent model mislocates abilities by up to three tenths of the field's spread (the pinned configuration: $n = 200$, thirty clusters, within-cluster correlation 0.65; maximum mislocation 2.49 against a spread of 8.24, median 0.34).

6 Dense covariance

The fitting procedure, stated so it can be reproduced. Write $s_i = \sqrt{\Sigma_{ii}}$, $S = \mathrm{diag}(s)$ and $C = S^{-1}\Sigma S^{-1}$ throughout: every fit, residual and closing solve below is in Σ, and C appears only in the clustering distance of stage 3. The tempting alternative — standardize to a correlation matrix, fit there, restore scales — fails: with $\Sigma = SCS$, $S = \mathrm{diag}(s)$, the choice-relevant residual after restoration is $PS(C - M)SP$, not $P(C - M)P$, and P does not commute with an unequal S. At $n = 2$ with equicorrelated C ($\rho = 0.9$) and $s = (1, 2)$, the correlation-space quotient objective declares the fit $M = (1 - \rho)I$ exact while the restored model gets the only identified quantity, the difference variance, wrong by nearly a factor of three (0.5 against 1.4); the choice-irrelevant direction in correlation coordinates is $S^{-1}\mathbf{1}$, not $\mathbf{1}$. A single global rescaling for conditioning is harmless. The test suite pins the two-alternative example and an exact-grammar round trip with variances spanning two decades.

The stages, in order:

1. Global factors: the quotient fit of §5 (`factor_model_projected`), $k = 3$ throughout, returning (V, D_0) by alternation. Each half-step is optimal in its own block, so in exact arithmetic the identified objective descends monotonically to a coordinatewise-stationary point. Above $n = 800$ the V-step is subspace iteration rather than a full eigendecomposition and its subspace is only approximate, so descent is enforced instead of assumed: the objective is evaluated each sweep at $O(n^2)$, a sweep that fails to improve it is discarded, and the alternation returns the best iterate. The objective is nonconvex in the pair, so no global certificate is claimed, and the residual report exposes the value reached. Multistart dispersion is the natural diagnostic, but it has to be measured in the right coordinates, which is worth stating because the obvious choice is the wrong one. Across four ensembles at $n = 300$ with eight random diagonal starts spanning two decades, every start reaches the same objective to 4×10^{-16} relative and no random start ever beats the shipped one. On three of the four the fits agree in choices as well, to a total variation of 10^{-14}. On block equicorrelation they do not: the starts agree on the objective and on D to every digit, and still price races that differ by a total variation of 0.25. The cause is degeneracy rather than optimization. Centering the six-block matrix leaves a five-fold tied leading eigenvalue, so a rank-three fit selects an arbitrary three-dimensional subspace of a five-dimensional one; every selection is exactly optimal and they imply different races. Raising the rank to the multiplicity makes the fit exact and collapses the disagreement to 8×10^{-4}. So the diagnostic to run is dispersion of the priced race across starts, not dispersion of the objective, which is blind to precisely the case that matters; and when it fires, the remedy is rank rather than a better optimizer.

2. The working residual is then projected once, $R = P\,(\Sigma - VV' - \operatorname{diag} D_0)\,P$: the raw residual still contains the choice-irrelevant common component the quotient fit rightly ignored, and letting later stages chase it manufactures projected error (measurably: the raw-residual variant distorts an input already of the form $VV' + \operatorname{diag}(D)$ by 1.7×10^{-2} of total variation; the projected pipeline returns it at the 2×10^{-4} node-noise floor).

3. Blocks: average-linkage clustering on $\sqrt{(1 - C)/2}$ cut to 20 clusters; for each cluster, the leading eigenvector of the within-cluster block of R with its diagonal zeroed, one rank-one loading per cluster (clusters are disjoint).

4. Residual promotion: the top $m = 5$ eigencomponents of what remains of R, diagonal zeroed, appended as further factor columns; negative eigenvalues are not promoted; numerically dead columns are dropped.

5. The closing diagonal solves the identified problem's own normal equations, $(P \circ P)\, d = \operatorname{diag}(P(\Sigma - V_{\text{all}} V_{\text{all}}')P)$ subject to $d \geq \ell$, with ℓ a small multiple of the mean variance. The Gram is exactly $(1 - \frac{2}{n})I + \frac{1}{n^2}\mathbf{1}\mathbf{1}^\top$, so the constrained solve is closed form in $O(n)$: substituting $d = \ell\mathbf{1} + x$ with $x \geq 0$ leaves the same Gram with a shifted right-hand side, and the KKT conditions make the active set a threshold set, solved by water-filling. It is the constrained minimizer, not an unconstrained solve with a floor applied afterwards, which the Gram's off-diagonal coupling would leave suboptimal. At $n = 2$ the Gram is rank one, only $d_1 + d_2$ is identified, and the symmetric representative is returned.

6. A second candidate — a pure eigenfit at the same total rank, closed by the same diagonal solve — is computed, and whichever candidate leaves the smaller choice-relevant residual is kept, the pipeline winning ties: the greedy factor-plus-blocks allocation is the wrong shape for globally smooth covariance, and the kernel battery below measures the repair.

The constants k, the cluster count and m are fixed, not tuned per ensemble; a spectral rule for m was tested and rejected (it saturates its cap for marginal gains and does not repair the failure case below). What results is a factor-plus-blocks race priced in one deterministic call, and the pipeline is the shipped one-call (`fit_covariance`; `race_probabilities` accepts `cov=` directly), not a research script.

The shipped call also reports its own quality and warns on two distinct failure modes: a large choice-relevant residual (the fit could not hold the matrix), and a well-fitted matrix whose pricing needs a high-rank, sharp factor integral (the quadrature, not the fit, is then the binding constraint; both regimes appear in the kernel battery below).

At $n = 2000$ the one-call fit-and-price takes four seconds (0.6 of them the fit) against thirty-six for a 10^6-draw simulation and agrees with it at its noise floor on the resolved bulk. For the 82 percent of tail alternatives with zero simulation wins, agreement in that referee is unverifiable entry by entry, and the set is selected by the same simulation that would have to certify it, so a fixed-set binomial bound does not apply. Sample splitting does apply, and we run it. Fix the zero-win set on one half of the draws, then count the wins an independent second half places anywhere in that set: because the set is fixed before the second half is looked at, its count is binomial in the set's true mass and a Clopper–Pearson interval is valid. On a dense $n = 2000$ correlation with three global factors and a forty-block layer, priced in a wide field where 63.6 percent of alternatives take no win in the first million paths, the model puts 1.13×10^{-4} of mass on that set and the second million paths place 111 wins in it, a 95 percent interval of $[9.13 \times 10^{-5}, 1.34 \times 10^{-4}]$. The model's mass is inside it. Scored on the second half alone, so that no statistic is evaluated on the draws that selected its own target, full-vector total variation is 4.4×10^{-3} against a split-half noise floor of 5.4×10^{-3}: at the referee's own resolution. Tail exactness at small n still comes from the Botev referee, which prices individual orthant probabilities rather than counting wins.

Fitting $VV' + \mathrm{diag}(D)$ *to* $P\Sigma P$ is the wrong objective (§5). With the identified objective and projected residuals throughout, the ensemble study below ran a raw top-k eigenfit pipeline and the identified pipeline over twenty seeds per ensemble. On thirteen of fifteen named ensembles the arms are statistically indistinguishable and none favors the raw arm. On block equicorrelation the identified arm is eight times better at the median (2.5×10^{-3} against 2.0×10^{-2}), with a worst seed of 3.2×10^{-3} against the raw arm's 0.162: the raw eigenfit spends its rank on a near-common component, as the identification argument predicts.

Accuracy depends on the generating ensemble, so it is reported per named ensemble of the `randomcov` library (pinned at commit `0d27a51`), now over *twenty seeds* per ensemble at $n = 300$, each seed refereed by its own 10^6-path simulation (the committed `run_ensembles4.py` and `run_kernel4.py` regenerate everything). The TV columns are full-vector total variation against the empirical frequency vector, zeros included, so the referee's own counting noise and any unresolved tail mass contribute to the reported number. That makes the statistic upward biased in expectation, by convexity, and not a realization-wise upper bound: on a single seed sampling error can cancel part of the model error; only the per-entry column is restricted, to entries with at least 25 referee wins. TV is reported as median, ninth decile and worst over seeds; the last column is the median per-entry absolute error on those resolved entries.

Representative rows, identified pipeline:

ensemble	med TV	q90 TV	worst TV	med $\lvert p - \hat{p}\rvert$
block equicorrelation	2.5×10^{-3}	3.0×10^{-3}	3.2×10^{-3}	1.4×10^{-5}
factor + sparse links	2.9×10^{-3}	6.7×10^{-3}	1.6×10^{-2}	1.9×10^{-5}
sparse-precision graphical	4.5×10^{-3}	5.6×10^{-3}	6.0×10^{-3}	1.5×10^{-5}
Wishart	1.2×10^{-2}	1.4×10^{-2}	1.5×10^{-2}	2.6×10^{-5}
AR(1)	4.6×10^{-2}	8.4×10^{-2}	1.5×10^{-1}	7.0×10^{-5}
residuals (dense-strong)	6.8×10^{-2}	8.6×10^{-2}	9.2×10^{-2}	2.5×10^{-4}

The medians are stable in the seed: any single seed reproduces them to within its own noise. Eight further named ensembles (LKJ, onion, vine, elliptope walk, animals, Marchenko–Pastur and spiked spectra, hierarchical) fall between the third and fifth rows; Archakov–Hansen sits with AR(1) at 4.9×10^{-2}.

The kernel family is the failure case, and the stratified battery (kernel type $\times$ length scale $\times$ budget, twenty seeds per cell, unit-square inputs) separates two distinct mechanisms that a single "kernel" row had conflated:

kernel	length scale	med TV, m=5	m=12	m=5, 2^{14} nodes
RBF	0.08	3.4×10^{-2}	3.1×10^{-2}	3.3×10^{-2}
RBF	0.2	2.6×10^{-2}	2.6×10^{-2}	1.2×10^{-2}
RBF	0.4	1.6×10^{-2}	1.7×10^{-2}	8.2×10^{-3}
Matérn-$\frac{3}{2}$	0.08	2.9×10^{-2}	2.5×10^{-2}	2.4×10^{-2}
Matérn-$\frac{3}{2}$	0.2	2.5×10^{-2}	2.3×10^{-2}	1.6×10^{-2}
Matérn-$\frac{3}{2}$	0.4	2.1×10^{-2}	1.7×10^{-2}	1.2×10^{-2}

At short length scale the bottleneck is *representation*: correlation is locality, the rank needed grows with the number of correlation lengths in the domain, and accordingly extra rank helps ($3.4 \to 3.1 \times 10^{-2}$) while extra quadrature does not. At long length scale the fit is not the problem at all — a rank-27 eigenfit holds the RBF draw at length scale 0.4 to a projected residual of 10^{-3} — but pricing a well-fitted smooth kernel means a twenty-seven–dimensional *sharp* factor integral (the closing diagonal is nearly zero, so conditional races are nearly deterministic), and there the node budget is the binding constraint: quadrupling rank does nothing while 2^{14} nodes halve the error, and 2^{16} nodes reach 6×10^{-3} on the diagnostic draw. Stage 6 of the pipeline is what reaches this regime at all: the greedy factor-plus-blocks allocation leaves a projected residual of 0.14 on the same draw that the matched-rank eigenfit holds to 10^{-3}, a misallocation worth up to 7×10^{-2} of total variation and not a limit of the grammar. The dispatch conclusion stands: genuinely local covariance wants sequential conditioning with cheap sparse-precision draws — Matérn is exactly the family with the SPDE route (Lindgren et al., 2011) — near-singular smooth covariance wants either a larger node budget or plain simulation, and a production front door should dispatch on the structure it detects; the shipped call's warnings distinguish the two regimes.

7 Benchmarks

The structure-aware per-alternative comparator. The fairest rival is not GHK but this paper's own conditioning run the literature's way: for each alternative separately, integrate the conditional independent-race integrand over the same Gauss–Hermite factor nodes (the construction of Butler and Moffitt (1982); `mvtnorm`'s `lpRR/slpRR` are its Monte Carlo form). With the same nodes and the same lattice the two computations are algebraically identical, and the measurement confirms it: at $n = 200$, rank 2, the per-alternative version agrees with the shared field to total variation 3×10^{-15} and costs 601 times as much (22 s against 36 ms; the committed `bench.py bm`). The comparison isolates the paper's contribution exactly: not the conditioning, which is forty years old, but the field shared across alternatives.

Against the field. All-n choice probabilities under the rank-2 factor covariance family of §4.1, against a 2^{20}-point QMC reference. Tail is the maximum absolute log-error over reference probabilities in $[10^{-4}, 10^{-3}]$; the $n = 10$ draw has none. Frequency simulation is run at wall clock matched to the lattice and at ten times that budget; "zeros" counts tail alternatives that received no draws.

	$n = 10$		$n = 30$		
method	time	TV	time	TV	tail
lattice race	2.5 ms	1.8×10^{-4}	4.7 ms	8.3×10^{-4}	0.07
Genz, full vector	46 ms	1.8×10^{-4}	1.40 s	8.3×10^{-4}	0.07
frequency, matched time	—	3.8×10^{-3}	—	1.0×10^{-2}	0.45, 2 zeros
frequency, 10× time	—	9.2×10^{-4}	—	2.0×10^{-3}	0.21
Mendell–Elston	2.0 ms	1.8×10^{-3}	18 ms	3.2×10^{-2}	1.30
Clark-type	1.4 ms	1.1×10^{-2}	17 ms	1.6×10^{-2}	1.33

Genz and the lattice agree to the reference's noise and to each other. Of the 300× cost ratio at $n = 30$, the one-at-a-time interface supplies an unavoidable n-fold component; the remainder reflects the differing integral representations, tolerances and implementations, and we do not apportion it further. At the paper's headline scale the n-fold component alone is decisive: one probability at $n = 200$ costs 0.43 s at `maxpts` $= 2.5 \times 10^5$ (agreement with the lattice to four to five digits; six digits needs multi-second budgets — the committed `bench.py genz200`), so the full vector costs 86 s against 26 ms for the lattice, a factor of 3×10^3 before any accuracy matching. The sequential-conditioning family is the only entry whose error grows silently: its tail entries are off by $e^{1.3} \approx 3.7$ with nothing in the output to say so. The race's tail figure is the reference's own noise; lattice self-convergence places its error near 10^{-8}.

Referee agreement. A replay harness maps each benchmark probability to its difference orthant and prices it through the two field-standard R implementations. `mvtnorm`'s Genz–Bretz agrees with the lattice to within its own reported error bound in every family: 3.8×10^{-7} against a bound of 8.1×10^{-7} at $n = 10$, and 2.7×10^{-8} against 5.8×10^{-8} on a case whose smallest probability is 3×10^{-10}. Botev's minimax tilting agrees to its own Monte Carlo noise, which in relative terms is 2.2×10^{-4} at $p \approx 3 \times 10^{-10}$: the tail claim rests on the named accuracy reference, not only on self-convergence. On the independent family an adaptive-quadrature referee reaches machine precision and the lattice matches it to 1.9×10^{-15}. An invariance battery (two-runner closed form, translation, permutation, symmetry) holds to 10^{-16} throughout, except the two-runner comparison at 7×10^{-9}, which is the lattice discretization itself.

Stress boundaries. An adversarial battery pushes each failure axis until something gives. Depth: against pure-relative adaptive quadrature, relative error is 4×10^{-13} at $p \approx 10^{-30}$

and 2×10^{-8} at $p \approx 10^{-50}$, with the first genuine breakdown near $p \approx 10^{-119}$, a laggard twenty-five standard deviations behind. Inversion: the round trip $p \to \mu \to p$ holds 10^{-9} in log even when the target contains a 10^{-10} entry or ties at the 10^{-9} level. The Gumbel base reproduces softmax to 10^{-15} at $n = 1000$; exact duplicates split exactly, and an ϵ perturbation moves the split linearly. The one operative boundary is factor strength. When loadings make implied correlations exceed roughly 0.9, the argmax integrand loses smoothness in factor space and Gauss–Hermite converges slowly in the node count: at loading scale 4 the 25-node rule still sits at 8×10^{-3} total variation. Scrambled-Sobol factor nodes restore reference-level accuracy (4×10^{-4}) at the same call: heavy correlation wants quasi-Monte-Carlo nodes, not deeper polynomial rules. The default now escalates the family automatically past sharpness 3, in both the Python and R implementations (scrambled Sobol and Halton respectively), which agree to 5×10^{-4} on the worst case; the figures above are the pre-escalation diagnosis.

Against GHK. All-n choice probabilities under a rank-2 factor covariance, against a 2^{20}-point QMC reference. TV is a bulk metric and blind to the tails, so the same tail column as above is reported: the maximum absolute log-error over reference probabilities in $[10^{-4}, 10^{-3}]$, which for probabilities this small is the log-odds error, and which the reference's own counting noise floors near 0.1.

	lattice race			GHK, $R = 10^3$			GHK, $R = 10^4$		
n	time	TV	tail	time	TV	tail	time	TV	tail
10	2.3 ms	1.8×10^{-4}	—	1.0 ms	8.4×10^{-3}	—	8 ms	2.1×10^{-3}	—
50	7.0 ms	1.7×10^{-3}	0.11	21 ms	4.2×10^{-2}	1.10	214 ms	1.2×10^{-2}	0.29
200	25 ms	2.4×10^{-3}	0.15	436 ms	3.2×10^{-2}	1.50	4.8 s	1.2×10^{-2}	0.35

The $n = 10$ draw has no reference probability in the tail band. The lattice's tail figures equal the reference's own noise; GHK's are two to ten times larger in log terms even at ten times the draws, which is the tail blindness the TV column cannot show.

Self-convergence places the lattice error near 10^{-8}. GHK error contracts as $O(R^{-1/2})$ and its measured complete-vector cost grows superquadratically, with exponent 2.15 to 2.8 across the measured range. The lattice is deterministic, supplies smooth matrix-free derivatives free of simulation noise (simulated likelihoods can be differentiated, but inherit the noise), and prices the tails. The claim: for complete-vector pricing and inversion under the factor, block, nested and tree covariance grammars, the shared-field construction removes the per-alternative repetition intrinsic to standard GHK implementations and dominates the per-alternative GHK implementation tested here in both accuracy and cost. GHK implementations with antithetics, quasi-random sequences or reordering heuristics would move the constants, not the n-fold shape. Estimation pipelines built around GHK are a larger object than this loop, and the estimation paragraphs below report where simulation remains competitive at face value. GHK also retains general rectangle probabilities and locality-structured covariance, and Botev (2017) remains the reference for single extreme tail probabilities.

Share inversion under sampling noise. The first estimation exercise is deliberately the saturated case. Simulate $T = 50{,}000$ choices from one fixed menu of $n = 30$ alternatives under a known rank-2 factor covariance and estimate the mean-zero utilities. With the menu fixed the model is saturated over the simplex interior, so the exact-gradient maximizer is the inversion of the observed shares and the exercise measures how faithfully each method performs that inversion under multinomial noise.

Each count is smoothed by one half before any arm sees it, which is the posterior-mean share vector under the multinomial Jeffreys prior, $\mathbb{E}[p_i \mid c] = (c_i + \frac{1}{2})/(T + n/2)$. (It is not the Jeffreys MAP: the Dirichlet($\frac{1}{2}$) posterior has exponents $c_i - \frac{1}{2}$, whose mode leaves empty cells on the boundary. Read as a penalized likelihood in simplex coordinates, add-half is the Dirichlet($\frac{3}{2}$) mode.) In the saturated model the estimator is the race inverse of that smoothed vector, and the simulated arms maximize the same add-half-smoothed criterion, so the comparison stays like for like. This is not cosmetic. The unpenalized saturated likelihood has *no* finite maximizer when any count is zero, which Theorem 1 states and this design would otherwise walk into: the smallest alternative here has $p = 2.5 \times 10^{-5}$, so its expected count is 1.27 and it is empty with probability $e^{-1.27} = 0.28$. Thirteen of the forty replications do have an empty cell, against the 11.2 that rate predicts, and on those an unpenalized optimizer returns a tolerance-dependent point on a boundary sequence rather than a maximizer. Forty replications, with the arms paired by construction since every method sees the same draw:

estimator	RMSE (median)	RMSE (mean ± s.e.)	median fit time
exact-gradient smoothed inversion	0.0168	0.0168 ± 0.0006	2.4 s
MSL, $R = 100$	0.0286	0.0280 ± 0.0006	3.5 s
MSL, $R = 1000$	0.0184	0.0185 ± 0.0005	39.9 s

The exact path is both faster and more accurate than the $R = 100$ simulator here, and beats the $R = 1000$ accuracy in a sixteenth of its time. Because the arms share every draw, the paired differences are the sharper statement: against $R = 100$ the exact arm is better by 0.0112 ± 0.0006, nearly nineteen standard errors, winning 39 of 40 replications; against $R = 1000$ by 0.0017 ± 0.0002, seven standard errors, winning 33 of 40. The $R = 1000$ margin is small in absolute terms, which is the honest characterization: enough simulation draws do approach the exact answer, at forty times the cost per fit, and the remaining gap is simulation bias rather than noise, since it survives pairing. The timing column is from an unloaded machine at the original eight replications; the accuracy columns are from all forty and are insensitive to load.

Covariate estimation. The overidentified case behaves differently, and we report it as we found it. Draw observation-specific design matrices X_t ($n = 10$ alternatives, $d = 3$ covariates, $T = 800$ observations), utilities $\mu_t = X_t\beta$, one choice per observation, covariance known, and estimate β by individual-level maximum likelihood: the exact path uses lattice probabilities with the analytic score (one Jacobian row per observation, a single field pass); the simulated path uses this paper's own Rust GHK with common random numbers and finite-difference gradients. Eight replications, means with standard errors:

estimator	RMSE($\hat{\beta}$)	median fit time
exact gradient MLE	0.0363 ± 0.0066	41.3 s
MSL, $R = 100$	0.0380 ± 0.0073	5.4 s
MSL, $R = 1000$	0.0367 ± 0.0067	39.6 s

All three estimators reach the statistical floor: with three parameters and eight hundred observations the simulation error averages out, and a well-implemented simulator at $R = 100$ is the fastest of the three at this size. The exact path's time is dominated by per-observation Python overhead (the compiled kernel accounts for under a second of the 41); the comparison at $n = 10$ measures plumbing, not method. What the exact likelihood buys in estimation is not this row of the table: it is freedom from the choice of R, gradients without simulation noise, the cost law as n grows, and diagnostics that simulation smooths over. The last point is not hypothetical: on the classic Fishing data, the unrestricted-covariance MNP likelihood evaluated exactly is boundary-seeking (still rising at loading norms near 4000), a ridge that GHK's simulation noise had been regularizing by accident. The behavior is reproducible from the committed estimator (`winning.mnprobit`), which detects the ridge and reports it as a `boundary_` diagnostic rather than returning the interior point simulation would have manufactured. For interpretability of that norm: the parameterization is rank-2 loadings with the reference alternative's rows fixed at zero and unit idiosyncratic variances, so global scale is pinned by the idiosyncratic normalization and the growing norm is a genuine choice-relevant direction, not the scale gauge; a profile likelihood along that normalized direction is the diagnostic a full treatment would add.

The Genz–Bretz score. Our quadrature comparator used `pmvnorm`, which exposes no score, with finite-difference gradients (31 vector evaluations per step at $n = 30$, roughly eight times the entire exact-gradient share-inversion fit). The `mvtnorm` manual (v1.4-2) also documents `slpmvnorm` and, for the reduced-rank covariance $BB' + \mathrm{diag}(D)$ of this paper's benchmarks, `lpRR` and `slpRR`: simulated log-likelihoods and score functions that integrate over the K factor dimensions. That interface is itself the conditioning construction of §4.1 with Monte Carlo in place of quadrature and no shared field across alternatives, which makes it the natural structure-aware comparator for a fuller estimation study.

GHK protocol. The GHK figures throughout use this paper's own Rust implementation: per-alternative sequential conditioning on the reference-differenced covariance with a fresh Cholesky factorization per alternative, pseudorandom draws (xoshiro family) with a per-alternative seed offset so that parameter evaluations share common random numbers, no antithetics, no quasi-random sequence, no variable-reordering heuristic, parallelized across alternatives. Cost figures are complete-vector; the large-n columns of the introduction are extrapolations of the measured law and are displayed as such.

Scale. Ten million contestants price in 245 seconds under the block grammar (ten thousand clusters, 257-point lattice, the streaming field pass; the committed `bench.py tenmillion`), and the Rust classic-calibration kernel runs 232 times the speed of the pure-Python implementation. Throughput at this size is a systems statement; the accuracy evidence lives at the scales the referees above can reach.

Parity. One vector file embeds the inputs and outputs of 22 scenarios spanning every claim above. The four language implementations of §8 replay them: twenty at machine precision, the optimizer-mediated scenarios to 10^{-6}.

8 Availability

`pip install winning` is pure Python; `winning[fast]` adds the Rust kernels as one abi3 wheel per platform. A base-R package mirrors the full interface, and a zero-dependency browser port drives a live demonstration at `winning.microprediction.org`.

References

J. H. Albert, S. Chib. Bayesian analysis of binary and polychotomous response data. *JASA* 88:669–679, 1993.

S. Berry. Estimating discrete-choice models of product differentiation. *RAND Journal of Economics* 25(2):242–262, 1994.

J. S. Butler and R. Moffitt. A computationally efficient quadrature procedure for the one-factor multinomial probit model. *Econometrica* 50(3):761–764, 1982.

Z. I. Botev. The normal law under linear restrictions: simulation and estimation via minimax tilting. *Journal of the Royal Statistical Society B* 79(1):125–148, 2017.

A. Börsch-Supan, V. Hajivassiliou. Smooth unbiased multivariate probability simulators for maximum likelihood estimation of limited dependent variable models. *Journal of Econometrics* 58:347–368, 1993.

M. Batram, D. Bauer. On consistency of the MACML approach to discrete choice modelling. *Journal of Choice Modelling* 30:1–16, 2019.

D. Bauer. *Rprobit: R package for the estimation of multinomial probit models.* `github.com/dbauer72/Rprobit`.

C. R. Bhat. The maximum approximate composite marginal likelihood (MACML) estimation of multinomial probit-based unordered response choice models. *Transportation Research Part B* 45(7):923–939, 2011.

L. Cappellari, S. P. Jenkins. Multivariate probit regression using simulated maximum likelihood. *Stata Journal* 3(3):278–294, 2003.

K. Chiong, A. Galichon, M. Shum. Duality in dynamic discrete-choice models. *Quantitative Economics* 7(1):83–115, 2016.

C. E. Clark. The greatest of a finite set of random variables. *Operations Research* 9:145–162, 1961.

Y. Croissant. Estimation of random utility models in R: the mlogit package. *Journal of Statistical Software* 95(11):1–41, 2020.

P. Cotton. Inferring relative ability from winning probability in multi-entrant contests. *SIAM Journal on Financial Mathematics* 12(1):295–317, 2021.

A. Galichon, B. Salanié. Cupid's invisible hand: social surplus and identification in matching models. *Review of Economic Studies* 89(5):2600–2629, 2022.

R. Gates. A Mata Geweke–Hajivassiliou–Keane multivariate normal simulator. *Stata Journal* 6(2):190–213, 2006.

A. Genz. Numerical computation of multivariate normal probabilities. *Journal of Computational and Graphical Statistics* 1:141–149, 1992.

J. Geweke. Bayesian inference in econometric models using Monte Carlo integration. *Econometrica* 57:1317–1339, 1989.

V. Hajivassiliou, D. McFadden, P. Ruud. Simulation of multivariate normal rectangle probabilities and their derivatives. *Journal of Econometrics* 72:85–134, 1996.

K. Imai, D. A. van Dyk. MNP: R package for fitting the multinomial probit model. *Journal of Statistical Software* 14(3):1–32, 2005.

M. P. Keane. A computationally practical simulation estimator for panel data. *Econometrica* 62:95–116, 1994.

S. Lerman, C. Manski. On the use of simulated frequencies to approximate choice probabilities. In *Structural Analysis of Discrete Data*, MIT Press, 1981.

L. Li. A general method for demand inversion. arXiv:1802.04444 (2018).

F. Lindgren, H. Rue, J. Lindström. An explicit link between Gaussian fields and Gaussian Markov random fields. *Journal of the Royal Statistical Society B* 73(4):423–498, 2011.

R. Loaiza-Maya, D. Nibbering. Scalable Bayesian estimation in the multinomial probit model. *Journal of Business & Economic Statistics* 40(4):1678–1690, 2022.

M. López de Prado. Building diversified portfolios that outperform out of sample. *Journal of Portfolio Management* 42(4):59–69, 2016.

D. McFadden. Econometric models of probabilistic choice. In C. Manski, D. McFadden, eds., *Structural Analysis of Discrete Data*, MIT Press, 1981.

R. McCulloch, P. Rossi. An exact likelihood analysis of the multinomial probit model. *Journal of Econometrics* 64:207–240, 1994.

N. Mendell, R. Elston. Multifactorial qualitative traits: genetic analysis and prediction of recurrence risks. *Biometrics* 30:41–57, 1974.

A. R. Solow. A method for approximating multivariate normal orthant probabilities. *Journal of Statistical Computation and Simulation* 37:225–229, 1990.

S. Stern. A method for smoothing simulated moments of discrete probabilities in multinomial probit models. *Econometrica* 60:943–952, 1992.

StataCorp. *Stata Choice Models Reference Manual: cmmprobit, asmprobit*. Stata Press, College Station, 2023.

L. L. Thurstone. A law of comparative judgment. *Psychological Review* 34:273–286, 1927.

K. Train. *Discrete Choice Methods with Simulation*. 2nd ed., Cambridge, 2009.

A Extrapolated GHK cost brackets

The lattice column below is measured; the GHK column is an extrapolation of the two measured exponents (2.15 and 2.8) far beyond any regime in which GHK has been run, and cache effects, parallelism, Cholesky reuse or a change of regime would move it.

n	lattice, measured	GHK at 10^4 draws, cost-law bracket	multiplier
10^4	0.18 s	20 hours – 3.7 days	$4 \times 10^5 - 2 \times 10^6$
10^5	2.7 s	0.3 – 6.5 years	$4 \times 10^6 - 8 \times 10^7$
10^6	29 s	46 – 4,100 years	$5 \times 10^7 - 4 \times 10^9$

Nobody prices a million-alternative field with GHK; the brackets say what the measured law implies about why. The ten-million forward-field benchmark of §7, which is actually run, is the meaningful systems demonstration.